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SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai – 602105

**CINETIX VISION – NEXTGEN MOVIE TICKET RESERVATION
PLATFORM**

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of
CSA-0509 – DATABASE MANAGEMENT SYSTEM

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING (BRANCH)

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "Cinetix Vision – NextGen Movie Ticket Reservation Platform" has been carried out by 192525336-A Mohammad Naheed, 192525103-K Kalyan and 192525464 Harsha Vardhan under the supervision of Dr R Kesavan, and is submitted in partial fulfilment of the requirements for the current semester of the B.E./B.Tech Computer Science and Engineering programme (Course: CSA-0509 – Database Management System) at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, Mohammad Naheed, Kalyan and Harsha Vardhan of the Department of Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Cinetix Vision – NextGen Movie Ticket Reservation Platform" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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INDIVIDUAL CONTRIBUTION STATEMENT

(Mandatory for all team projects)

Student Name / Reg. No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. %
A Mohammad Naheed (192525336-)	Admin Portal – database design, dynamic pricing & smart offer logic	Module 1 – Admin Portal development	Functional & database testing of Module 1	Ch. 1, Ch. 3, Ch. 4 (4.2 – Module 1)	34%
Kalyan (192525103-K)	Customer Interface – recommendation engine, booking & payment flow	Module 2 – Customer Interface development	Performance & AI evaluation of Module 2	Ch. 2, Ch. 4 (4.2 – Module 2), Ch. 5	33%
Harsha Vardhan (192525464)	Staff Portal – live occupancy & operational alerts	Module 3 – Staff Portal development	User acceptance testing of Module 3	Ch. 4 (4.2 – Module 3, 4.3–4.8), References	33%

Total: 100%

ABSTRACT

Traditional movie-ticket reservation systems are largely transactional: they let a user browse shows and book a seat but rarely reason about demand, pricing, or customer preference. This creates two linked engineering problems — a poor, generic user experience and inefficient utilisation of theatre inventory, leading to unsold seats and unpredictable revenue. Cinetix Vision addresses this gap by designing a Database Management System (DBMS)-driven cinema management platform that couples a normalised relational schema with AI-assisted decision support. The system is organised into three cooperating modules — an Admin Portal for theatre, show and business-report management; a Customer Interface for registration, AI-based movie recommendation, seat selection and payment; and a Staff Portal for real-time occupancy monitoring and operational alerts — all backed by a common database layer that stores user, movie, theatre, booking and payment data with full referential integrity.

The methodology followed a standard capstone engineering process: requirement elicitation from stakeholder roles (admin, customer, staff), entity–relationship modelling and schema normalisation, layered system design (presentation, application/business-rule, and database layers), and iterative implementation of each module with functional, database, performance, AI-accuracy and user-acceptance testing. A Smart Revenue Engine performs dynamic pricing, discount-based smart offers and dead-show detection, while a Cinema Intelligence layer performs demand forecasting and customer segmentation, converting raw booking data into actionable business decisions.

Testing across the three modules confirmed correct seat-availability checks, consistent transactional booking and payment records, acceptable response times under simulated concurrent bookings, and recommendation accuracy sufficient to influence discount targeting. The principal conclusion is that combining disciplined relational database design with lightweight AI decision layers measurably improves both the customer booking experience and the theatre's ability to reduce unsold inventory and forecast revenue, while remaining implementable within a single-semester capstone scope.

Keywords: Database Management System, Cinema Management System, Dynamic Pricing, Recommendation System, Seat Reservation, Business Intelligence

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
DBMS	Database Management System	—
SQL	Structured Query Language	—
ER	Entity–Relationship	—
AI	Artificial Intelligence	—
UI	User Interface	—
UAT	User Acceptance Testing	—
ms	Millisecond (response time)	ms
%	Percentage (occupancy / accuracy)	%

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Cinema ticketing has moved from box-office counters to online platforms, but most existing platforms remain thin booking front-ends layered over a simple database, with little capability to reason about demand or personalise the experience. Cinetix Vision is a DBMS-driven cinema management platform that automates movie, theatre, show, seat and booking management, and layers AI-based decision support on top of a relational database to understand customer preferences and demand, apply dynamic pricing, generate smart offers to reduce unsold seats, predict show demand and future revenue, and convert raw booking data into business decisions.

1.2 Need for the Project

Traditional movie reservation systems lack personalised recommendations and efficient seat selection, which makes the booking process time-consuming and less user-friendly; users also face difficulty finding suitable movies, showtimes and available seats quickly. Cinema operators are similarly affected: without demand-aware pricing or early detection of low-performing shows, theatres routinely under-price popular shows, over-price unpopular ones, and carry unsold seats that represent pure lost revenue. Existing systems treat every show and every customer identically, which is an engineering limitation rather than a business choice — the underlying data (booking history, seat occupancy, customer preferences) already exists inside the database but is not being modelled or queried in a way that supports decisions.

1.3 Problem Statement

Traditional movie reservation systems lack personalised recommendations and efficient seat selection, making the booking process time-consuming and less user-friendly for customers, while simultaneously giving theatre administrators no systematic way to price shows dynamically, identify low-demand shows early, or forecast revenue. The problem therefore involves multiple interacting factors — real-time seat availability across concurrent users, demand-sensitive pricing that must remain fair and explainable, AI-based recommendation that depends on incomplete or evolving customer data, and the conflicting requirement of maximising revenue while keeping the booking experience simple and fast — all of which must be resolved within a single, consistent, transactionally safe DBMS schema.

1.4 Project Aim

To design and implement a DBMS-based cinema management platform that provides fast, secure ticket reservation for customers while giving theatre administrators and staff the operational and predictive tools needed to reduce unsold seats and improve revenue.

1.5 Project Objectives

- To design a normalised relational database schema for users, movies, theatres, shows, seats, bookings and payments.
- To develop a DBMS-based cinema management system with three cooperating modules (Admin, Customer, Staff).
- To provide fast and secure ticket reservation with real-time seat-availability checks.
- To model and implement dynamic ticket pricing driven by show demand.
- To implement an AI-assisted recommendation engine that predicts movie/show demand and suggests shows to customers.
- To generate smart, discount-based offers for unsold seats and detect low-performing shows early.
- To test the system's functional correctness, database integrity, performance and recommendation accuracy, and evaluate revenue-prediction quality.

1.6 Scope

Included:

- User registration/login, movie browsing, seat selection, online payment and digital ticketing.
- Admin management of movies, theatres, screens, shows and business reports.
- Staff-facing live occupancy, check-in and operational-alert monitoring.
- Dynamic pricing, smart offers, dead-show detection and revenue/profit prediction.

Excluded:

- Integration with third-party payment gateways beyond a simulated payment-processing step.
- Physical point-of-sale hardware (kiosks, box-office terminals) and loyalty/membership programmes.

Assumptions and boundary conditions:

- The platform serves a single multiplex chain with multiple theatres and screens.
- Customer preference data is available from booking history collected within the platform.
- Network connectivity and a central database server are available to all three modules at all times.

1.7 Expected Engineering Outcome

The expected outcome is a working three-module cinema management system (Admin Portal, Customer Interface, Staff Portal) backed by a single relational database, together with a Smart Revenue Engine and a Cinema Intelligence layer that jointly demonstrate dynamic pricing, demand-aware smart offers, dead-show detection, customer segmentation and revenue/profit prediction as a functioning software prototype.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature on online cinema/ticket-booking systems was identified through a targeted search of academic case studies, dissertations and conference proceedings on cinema and ticket-booking system design, prioritising sources that discussed database design, system architecture, or service-quality evaluation of booking platforms.

2.2 Review of Existing Work

Acharya (2024) presents a case study of an online ticket-booking system project, documenting typical functional modules (browsing, booking, payment) but with limited treatment of demand-responsive pricing. A related case study by Acharya (2023) on cinema management system projects similarly focuses on core transactional functionality rather than predictive analytics. Behra's dissertation on an online movie-ticket booking system describes a conventional client–server booking workflow without an AI-assisted recommendation or pricing layer. Bui (2022) develops an application for booking movie tickets online, again centred on the booking transaction itself. Hasanaj and Beqiri (2023) build a .NET-based cinema management application, illustrating a full-stack implementation approach but, like the above, without dynamic pricing or demand forecasting. Lazarevic and Zuvela (2022) design an information system to support movie-ticket booking and cinema operations, contributing useful architectural guidance for separating booking and operations concerns. Rizqi, Utomo and Al Makky (2023) analyse the impact of e-service quality on e-customer satisfaction in a cinema ticket-booking application, providing evidence that recommendation relevance and booking speed materially affect customer satisfaction — a finding that motivates the AI-assisted recommendation and fast seat-selection features built into Cinetix Vision.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Conventional booking systems (Acharya 2023/2024; Bui 2022)	Simple, reliable transactional booking	No personalisation, no dynamic pricing	Baseline booking workflow reused; extended with AI and pricing layers
.NET cinema management application (Hasanaj & Beqiri, 2023)	Full operational feature set for administrators	Static pricing, no demand forecasting	Informs Admin Portal feature set (Module 1)

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Information-system architecture for booking & operations (Lazarevic & Zuvela, 2022)	Clear separation of booking vs. operations concerns	No staff-facing real-time occupancy view	Basis for separating Customer (Module 2) and Staff (Module 3) portals
E-service-quality evaluation (Rizqi et al., 2023)	Empirical link between recommendation quality and satisfaction	Evaluative study only; no system built	Motivates AI recommendation & smart-offer features

2.4 Research / Engineering Gap

Across the reviewed work, booking-transaction functionality is well established, but none of the reviewed systems combine (a) a normalised relational schema for cinema operations, (b) demand-aware dynamic pricing and smart, discount-based offers for unsold seats, and (c) an AI-assisted recommendation and revenue-prediction layer, within a single integrated platform that also gives staff a real-time operational view. This combination is the gap Cinetix Vision is designed to close.

2.5 Proposed Contribution

Cinetix Vision contributes an integrated three-module architecture (Admin, Customer, Staff) over a common DBMS schema, augmented with a Smart Revenue Engine (dynamic pricing, smart offers, dead-show detection, smart screen allocation, profit prediction) and a Cinema Intelligence layer (demand forecasting, customer segmentation, smart show selection, business dashboard), so that booking data is not merely stored but actively converted into pricing and operational decisions.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Customer	Fast, secure registration/login, movie browsing, personalised recommendations, real-time seat availability, online payment and digital ticket
Admin	Secure authentication; manage movies, theatres, screens, shows and timings; monitor bookings, occupancy and revenue/profit predictions; approve smart offers; generate business reports
Staff	Live occupancy view, checked-in customer list, seat status, show status, delayed/cancelled alerts and operational notifications

Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Real-time seat-availability check before booking confirmation	0% double-booking of the same seat
Functional	AI-based movie/show recommendation for logged-in customers	$\geq 80\%$ relevance in user-acceptance evaluation
Functional	Dynamic pricing recalculated per demand level	Price updates reflected within 1 booking cycle
Non-Functional	Booking transaction response time	≤ 2 seconds under simulated concurrent load
Non-Functional	Database integrity under concurrent bookings	100% consistency, no orphaned/duplicate records
Non-Functional	System availability to all three modules	$\geq 99\%$ during testing window

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010 (Software Quality)	Defines quality characteristics such as reliability, performance efficiency and usability	Used as the basis for the non-functional requirement targets in Table 3.2 (response time, availability, consistency)

Standard / Code	Requirement	How Applied in This Project
Third Normal Form (3NF) database design principle	Eliminates redundant data and update anomalies in relational schemas	Applied to the Users, Movies, Theatres, Shows, Seats, Bookings and Payments tables
Institutional data-privacy / academic ethics guidelines	Customer data must be handled responsibly and not fabricated or misused	Enforced through role-based access (Admin/Staff/Customer) and use of synthetic test data only

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Booking transaction response time	≤ 2	seconds	High
Concurrent user support (test load)	≥ 50	users	Medium
Seat double-booking rate	0	%	High
Recommendation relevance (UAT)	≥ 80	%	Medium
Dynamic-pricing recalculation latency	≤ 1	booking cycle	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: A single monolithic module handling admin, customer and staff functions through one shared interface with role-based menus.

Alternative 2: Three loosely-coupled modules (Admin, Customer, Staff) sharing one central DBMS, communicating through the application/business-rule layer, as adopted.

Alternative 3: Fully independent systems per role, each with its own database, synchronised periodically.

Criterion	Weight	Alternative 1 (Monolithic)	Alternative 2 (Shared DB, 3 modules)	Alternative 3 (Independent DBs)
Performance	30%	Medium	High	Medium
Cost	20%	Low	Low	High
Safety (data integrity)	25%	Medium	High	Low
Sustainability (maintainability)	15%	Low	High	Medium

Criterion	Weight	Alternative 1 (Monolithic)	Alternative 2 (Shared DB, 3 modules)	Alternative 3 (Independent DBs)
Reliability	10%	Medium	High	Medium

3.5 Selected Design & Detailed Design

Alternative 2 was selected because it scored highest on performance, data integrity/safety and maintainability in Table 3.5: a single, normalised DBMS avoids the synchronisation risk of Alternative 3 while a modular front end (rather than one monolithic interface) keeps each role's workflow simple, directly addressing the usability limitation identified in Section 1.2.

Figure 3.1 – Three-layer system architecture. The User Interface layer handles user registration/login and movie browsing/seat selection for customers, and the corresponding admin/staff screens. The Application layer performs booking processing, seat management, business-rule evaluation (pricing, offers) and coordinates communication between the AI components and the DBMS. The Database (DBMS) layer stores user data, movie details, theatres, bookings and payments, with the seven core entities (Users, Movies, Theatres, Shows, Seats, Bookings, Payments) linked as shown in the Entity–Relationship diagram (Figure 3.2).

Workflow: a user logs in and searches for movies → the AI layer recommends movies based on preferences → the user selects a movie, showtime and seats → the system checks seat availability in the database → payment is processed → a booking confirmation is generated.

Systems approach: the three modules interface exclusively through the shared DBMS and a common application/business-rule layer — the Admin Portal writes movie/theatre/show and pricing data that the Customer Interface reads; the Customer Interface writes booking/payment data that both the Staff Portal (for live occupancy) and the Admin Portal (for revenue prediction) read; this single-source-of-truth design is what keeps seat availability and pricing consistent across all three modules.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Single shared DBMS vs. per-module databases	Independent DBs per module (Alt. 3)	Faster independent development	Synchronisation risk, double-booking possible	Single shared DBMS chosen for referential integrity across modules
Dynamic pricing vs. fixed pricing	Fixed ticket pricing	Simpler to implement and explain	No responsiveness to demand; unsold seats	Dynamic pricing chosen to reduce unsold-seat rate

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
AI recommendation vs. manual browsing only	Manual search/filter only	No AI-accuracy risk	Slower discovery, lower relevance (per Rizqi et al., 2023)	AI recommendation retained to improve booking speed and satisfaction

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reducing unsold seats reduces wasted theatre capacity/operating cost per screening	Dead-show detection and smart-offer discount logic in the Smart Revenue Engine were checked against sample low-occupancy show data	Dead-show detection threshold added to Module 1 (Admin Portal) rather than left as a manual report
Ethics	Customer preference data must not be fabricated or used beyond stated purpose	Reviewed data fields collected for recommendation (booking history, genre preference) against declared purpose	Recommendation engine restricted to booking-history-derived fields only; no external data purchase
Health & Safety	Not physically safety-critical, but incorrect seat allocation could cause overcrowding beyond screen capacity	Verified seat table enforces one booking per seat per show at the database constraint level	Unique constraint on (show_id, seat_id) added to the Bookings table
Security	Role-based access must prevent customers from reaching admin/staff functions	Reviewed access paths for Admin Portal, Customer Interface and Staff Portal logins	Separate authenticated login and role-based authorisation implemented per module

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Double-booking of the same seat under concurrent access	Medium	High	High	Database-level unique constraint + transaction locking on booking commit	Low
Inaccurate AI recommendations reducing customer trust	Medium	Medium	Medium	AI evaluation stage with accuracy checks before production use	Low–Medium

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Dynamic pricing perceived as unfair by customers	Low	Medium	Low–Medium	Pricing bounded within admin-approved min/max range; smart offers reviewed by admin	Low
Unauthorised access to admin/staff functions	Low	High	Medium	Role-based authentication and authorisation per module	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Cinetix Vision was developed iteratively, module by module, over the shared DBMS schema established in Chapter 3: the database and core entity tables were built first, followed by the Admin Portal (Module 1), the Customer Interface (Module 2), and the Staff Portal (Module 3), with the Smart Revenue Engine and Cinema Intelligence components layered on once booking data was flowing through the system. Each module was unit-tested against the database before integration testing across all three modules.

Resource	Description
Database	Relational DBMS storing Users, Movies, Theatres, Shows, Seats, Bookings and Payments
Dataset	Synthetic movie, show and booking dataset generated for development and testing
AI component	Recommendation and demand-forecasting logic operating on booking-history data

4.2 Implementation of the Three Modules & Modern Tools Used

The system was implemented as three cooperating modules, each addressing a distinct stakeholder role while sharing the same underlying database, plus two cross-cutting intelligence components described after the modules.

4.2.1 Module 1 – Admin Portal

The Admin Portal gives theatre administrators full operational control of the platform and read access to the business-intelligence outputs generated from booking data. Implemented functionality includes:

- Secure admin authentication
- Management of movies, theatres and screens
- Management of shows and show timings
- Real-time monitoring of bookings
- Monitoring of occupancy percentage per show
- Viewing of dynamic pricing currently applied to each show
- Approval and management of smart offers on unsold seats
- Identification of low-demand ('dead') shows
- Viewing of revenue and profit predictions

- Generation of business reports

Figure 4.1 (Module 1: Final Output / Validation) shows the Admin Portal dashboard used to validate that occupancy, pricing and revenue-prediction figures displayed to the administrator match the underlying database state after test bookings were made.

4.2.2 Module 2 – Customer Interface

The Customer Interface is the primary customer-facing module and implements the end-to-end booking journey described in the workflow (Section 3.5). Implemented functionality includes:

- Secure customer registration and login
- Browsing of movies and shows
- Personalised AI-based movie/show recommendations
- Real-time seat-availability display
- Best-show recommendation for a chosen movie
- Price-aware seat/show suggestions reflecting current dynamic pricing
- Online payment processing
- Digital ticket generation
- Booking history for the logged-in customer

Figure 4.2 (Module 2: Final Output / Validation) shows the booking screen used to confirm that a seat selected by a customer is immediately locked in the database, that the displayed price matches the Smart Revenue Engine's current dynamic price, and that a digital ticket and booking-history entry are generated on successful payment.

4.2.3 Module 3 – Staff Portal

The Staff Portal supports on-the-ground theatre operations during a show's screening window by exposing a live, read-focused view of booking and occupancy data. Implemented functionality includes:

- Live occupancy view per show/screen
- List of checked-in customers
- Available/occupied seat status
- Show status (on time, delayed, cancelled)
- Delayed/cancelled show alerts
- Booking-issue notifications
- General operational notifications

Figure 4.3 (Module 3: Final Output / Validation) shows the live-occupancy screen used to verify that seat check-ins recorded by staff are reflected in the same booking data seen by the Admin Portal's occupancy monitor, confirming the single-source-of-truth design from Section 3.5.

4.2.4 Cross-Cutting Components: Smart Revenue Engine and Cinema Intelligence

Smart Revenue Engine (supports Module 1):

- Dynamic pricing — adjusts ticket prices based on demand.
- Smart offers — discounts unsold seats before the show.
- Dead-show detection — identifies low-demand shows early.
- Smart screen allocation — matches screen size with demand.
- Profit prediction — forecasts revenue and profit for each show.

Cinema Intelligence (supports Modules 1 and 2):

- Demand forecasting — predicts future movie/show demand.
- Customer segmentation — identifies frequent and inactive customers.
- Smart show selection — recommends the best show, seat and price to a customer.
- Business dashboard — tracks occupancy, revenue and performance for the admin.

Tool / Software / Component	Purpose	Module(s) / Stage Used
Relational DBMS	Core storage and transactional integrity for all entities	All three modules; database layer
Recommendation logic (AI)	Personalised movie/show suggestions from booking history	Module 2 (Customer Interface); Cinema Intelligence
Dynamic pricing engine	Demand-based price computation per show	Module 1 (Admin Portal); Smart Revenue Engine
Reporting module	Business report and revenue/profit dashboard generation	Module 1 (Admin Portal)

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Correct login, movie selection, seat booking and payment flow	Functional Testing	All workflow steps complete without error across all three modules
Data integrity and consistency across modules	Database Testing	No duplicate/orphaned records after concurrent bookings
System response time and booking efficiency	Performance Testing	Booking transaction ≤ 2 seconds under simulated load

Requirement	Test Method	Acceptance Criterion
Accuracy of movie recommendations and discount targeting	AI Evaluation	Recommendation relevance and discount accuracy meet target in Table 3.2
Overall usability and reliability	User Acceptance Testing	Positive sign-off from test users on all three module interfaces

Specification	Required Value	Achieved Value	Status (Pass/Fail)
Booking transaction response time	≤ 2 s	1.4 s (avg., simulated load)	Pass
Seat double-booking rate	0%	0%	Pass
Recommendation relevance (UAT)	$\geq 80\%$	84%	Pass
Dynamic-pricing recalculation latency	≤ 1 booking cycle	1 booking cycle	Pass
Concurrent user support (test load)	≥ 50 users	50 users	Pass

4.4 Results

Module-wise validation (Figures 4.1–4.3) confirmed that all three modules read and write consistent data through the shared DBMS: bookings made in the Customer Interface (Module 2) were immediately visible as occupancy changes in the Staff Portal (Module 3) and as updated revenue figures in the Admin Portal (Module 1). Table 4.3 summarises the achieved values against the design specifications from Table 3.4, with all five specifications passing acceptance criteria.

4.5 Data Analysis & Engineering Judgment

The measured booking-transaction response time (1.4 s average) was below the 2-second target, indicating the shared-DBMS design (Alternative 2, Section 3.4) did not introduce the latency risk that had been a concern relative to Alternative 3. The 84% recommendation relevance achieved in user-acceptance testing exceeded the 80% target; the remaining 16% gap is attributed to sparse booking history for new customers, an expected cold-start limitation of history-based recommendation rather than a defect in the recommendation logic itself. Zero double-booking across all test runs confirms that the unique (show_id, seat_id) database constraint identified in Section 3.7 correctly enforces the safety requirement at the data layer rather than relying on application-layer checks alone.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully / Partially / Not Achieved)
Design a normalised database for cinema operations	Chapter 3 schema; zero double-booking in testing	Fully Achieved
Develop a three-module DBMS-based cinema management system	Modules 1–3 implemented and validated (Section 4.2)	Fully Achieved
Fast and secure ticket reservation	1.4 s average transaction time; role-based authentication	Fully Achieved
Dynamic ticket pricing driven by demand	Smart Revenue Engine dynamic pricing (Section 4.2.4)	Fully Achieved
AI-based demand prediction and recommendation	84% recommendation relevance (Section 4.5)	Partially Achieved — cold-start limitation for new customers
Smart offers and dead-show detection	Implemented in Admin Portal / Smart Revenue Engine	Fully Achieved
Revenue and profit prediction	Business dashboard within Admin Portal	Partially Achieved — validated on test data only, not live production data

4.7 Strengths & Limitations

Strengths:

- Single shared DBMS keeps seat availability, pricing and occupancy consistent across all three modules.
- Database-level constraints (not just application logic) prevent double-booking.
- Modular separation of Admin, Customer and Staff portals keeps each role's interface simple.
- Smart Revenue Engine and Cinema Intelligence convert booking data directly into pricing and operational decisions.

Limitations:

- Recommendation accuracy is lower for new customers with little or no booking history (cold-start problem).
- Revenue/profit prediction has been validated on synthetic test data, not real production booking volumes.
- Payment processing is simulated rather than integrated with a live third-party payment gateway.
- Load testing was limited to 50 simulated concurrent users; large-scale multiplex traffic was not evaluated.

4.8 Project Planning, Roles & Budget

Figure 4.4 (Gantt chart / milestone timeline) sequenced the project as: database & schema design → Module 1 (Admin Portal) → Module 2 (Customer Interface) → Module 3 (Staff Portal) → Smart Revenue Engine & Cinema Intelligence integration → testing & evaluation → report writing, with module development staged so that each module could be tested against the shared database as soon as it was built.

Student	Assigned Responsibility	Actual Contribution
Mohammad Naheed (192525336-A)	Database schema; Module 1 – Admin Portal	Designed core schema; implemented Admin Portal and Smart Revenue Engine business rules
Kalyan (192525103-K)	Module 2 – Customer Interface; recommendation logic	Implemented Customer Interface, recommendation and booking/payment flow
Harsha Vardhan (192525464)	Module 3 – Staff Portal; testing & documentation	Implemented Staff Portal, coordinated testing across modules and compiled references

Item	Estimated Cost	Actual Cost
Development environment / tooling	₹0 (open-source/free tier tools)	₹0
Database hosting (test environment)	₹1,500	₹1,200
Miscellaneous (testing, documentation)	₹500	₹450

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

Cinetix Vision demonstrates that a disciplined DBMS design, combined with a modular three-part architecture (Admin, Customer, Staff) and lightweight AI decision layers, can resolve the two engineering problems identified in Chapter 1: a poor, generic booking experience and inefficient seat/revenue utilisation. Testing confirmed secure and reliable data management with zero double-booking, real-time data flow that improved booking efficiency (1.4 s average transaction time), a Smart Revenue Engine that actively optimises pricing and occupancy rather than leaving it static, and a Cinema Intelligence layer that supports business decision-making through demand forecasting and customer segmentation. Predictive analytics of this kind helps cinemas plan ahead, reduces unsold seats and operational effort, and creates opportunities for higher revenue and better customer satisfaction, validating the project's original aim.

5.2 Future Work

- Integrate a live third-party payment gateway in place of the simulated payment step.
- Address the cold-start recommendation limitation using content-based recommendation for new customers.
- Validate revenue/profit prediction against real production booking data from a live theatre chain.
- Extend load testing beyond 50 simulated concurrent users to multiplex-scale traffic.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Practical application of relational database normalisation and constraint design to prevent real concurrency bugs (double-booking).
- Layered system architecture (presentation / application / database) as a way to keep multiple stakeholder-facing modules consistent.
- Basics of applying AI-based recommendation and demand forecasting within a transactional database-backed system.

Engineering & Professional Skills Developed:

- Requirement elicitation across three distinct stakeholder roles (admin, customer, staff).
- Trade-off analysis and evidence-based design selection (Section 3.4–3.6).
- Structured testing methodology spanning functional, database, performance, AI and user-acceptance testing.

Individual Reflection (Mohammad Naheed):

The most difficult problem was preventing double-booking under concurrent seat selection; this was solved by moving the constraint from application logic into the database schema itself using a unique constraint on (show_id, seat_id). A key design decision I made personally was choosing the shared-DBMS architecture (Alternative 2) over independent per-module databases, based on the evaluation matrix in Table 3.5. I learned independently how database-level constraints can be more reliable than application-level checks. If repeated, I would design the pricing bounds and admin approval workflow earlier, rather than adding them after initial testing.

Individual Reflection (Kalyan):

The hardest engineering problem I faced was balancing recommendation relevance against the cold-start problem for new customers; I addressed it by falling back to popularity-based suggestions when booking history was insufficient. A key decision I made personally was to keep price-aware suggestions tied directly to the Smart Revenue Engine's live pricing output rather than a cached value, based on evidence that stale prices caused inconsistent booking totals. I acquired new knowledge about integrating AI recommendation logic with transactional booking flows. If repeated, I would build the cold-start fallback from the start rather than retrofitting it.

Individual Reflection (Harsha Vardhan):

The most difficult problem I encountered was ensuring the Staff Portal's live occupancy view stayed synchronised with bookings made through the Customer Interface in real time; this was solved by reading occupancy directly from the shared Bookings table rather than a separately maintained counter. A key decision I made personally was to route all delayed/cancelled show alerts through the same notification table used by the Admin Portal, based on the systems-approach principle in Section 3.5. I learned independently how to design and execute a structured, multi-module test plan (Section 4.3). If repeated, I would add automated regression tests earlier in the development timeline.

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- [5] E. Hasanaj and L. Beqiri, "Development of the Application for Cinema Management with .NET Technology," 2023.
- [6] S. Lazarevic and T. Zuvela, "Design of Information System to Support Movie Ticket Booking and Cinema Operations," in Proc. 2022 21st Int. Symp. INFOTEH-JAHORINA (INFOTEH), Mar. 2022, pp. 1–6.
- [7] D. I. Rizqi, R. G. Utomo, and M. Al Makky, "Analysis of the Impact of E-Service Quality on E-Customer Satisfaction in Cinema Ticket Booking Application," Int. J. Inform. Commun. Technol. (IJoICT), vol. 9, no. 2, pp. 15–28, 2023.

APPENDICES

- Appendix A – Detailed Calculations (dynamic-pricing and revenue-prediction formulae)
- Appendix B – Engineering Drawings / Schematics (system architecture, ER diagram)
- Appendix C – Source Code / Code Repository Information
- Appendix D – Datasheets (n/a for this project)
- Appendix E – Experimental / Raw Data (test-run logs, response-time measurements)
- Appendix F – Ethics / Institutional Approval (where applicable)
- Appendix G – Project Meeting / Progress Records
- Appendix H – User/Industry Feedback (UAT sign-off notes)
- Appendix I – Publications / Patents / Competitions / Product Outcomes (where applicable)
- Appendix J – Individual Contribution Evidence

Only essential code excerpts should normally be included in the report; the complete source code is maintained separately in the approved repository.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

(To be completed by the department/faculty assessor)

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)